Greetings

There are n people on a number line. The i-th person starts at position a_i and wants to travel to position b_i (their destination) in the positive direction ($a_i < b_i$). All starting positions and destinations ($2n$ numbers in total) are distinct.

Now, all n people start moving towards their destinations simultaneously, at a speed of 1 unit per second. They stop moving once they reach their destination. Two people *greet* each other if they occupy the same coordinate at the same moment. Note that a person can greet others even after they have stopped at their destination.

You are given n and the arrays `a` and `b`, which contain the starting positions and destinations of all n people, respectively. Compute the total number of greetings that will occur.

Example Test Cases:

Test Case 1
Input: $n = 2$, `a` = `[2,1]`, `b` = `[3,4]`
Output: 1
Explanation: The two people will meet at position 3 and greet each other.

Test Case 2
Input: $n = 6$, `a` = `[2,3,4,1,7,-2]`, `b` = `[6,9,5,8,10,100]`
Output: 9

Test Case 3
Input: $n = 4$, `a` = `[-10,-5,-12,-13]`, `b` = `[10,5,12,13]`
Output: 6

Expected Time Complexity: $O(n \log n)$

Input Format:
An integer n, and two arrays `a` and `b` of length n each.

Output Format:
Return a single integer, the total number of greetings that will occur.

Constraints:

1. $1 \leq n \leq 2 \cdot 10^5$
2. $-10^9 \leq a_i < b_i \leq 10^9$

3. For each test case, all $2n$ numbers $a_1, \ldots, a_n, b_1, \ldots, b_n$ are distinct.

4. The sum of n over all test cases does not exceed $2 \cdot 10^5$.

Instructions:

You have been provided with two template files, containing the `main` and `solve` functions respectively. The `main` function, which has been completed for you, parses the input for each test case. For each test case, it calls the `solve` function, which must compute the total number of greetings that will occur.

The `solve` function must compute this in $O(n \log n)$ time.

(Refer to the `Instructions.pdf` file for detailed guidelines on the input-output format and additional test cases.)

Library Usage

For this lab, you are permitted to use any modules which are part of the standard library of either `C++` or `python`. For `C++`, do not add any extra `#include` directives; the provided `bits/stdc++.h` covers all allowed modules.